Introduction to Course Project

ELG 5255 (Applied Machine Learning)

2026 Fall term

- **Project Overview:** The course project provides hands-on opportunities for you to apply machine learning (ML) concepts and algorithms you've learnt to a real-world engineering problem. **The project has two stages for reporting: 1)** A state-of-the-art review of applied ML algorithms in your selected topic and **2)** A code implementation of at least one applied machine learning algorithm in the selected topic.

- **Group Formulation:** Students can work in group sizes ranging from **two to four students**. Students are responsible for forming your own groups. Each group member will be evaluated separately based on their contribution to the course project and therefore, all deliverables **must clearly state the contribution of each group member,** and all group members **must sign to agree**.

- **Confirmation of Group Formulation (Using Google Docs):** Please use the following Google doc to fill in your confirmed group information. If you are looking for groups to join or more members, you can also use the Google Doc to find members. Detailed instructions are provided in Google Doc. **Confirmation deadline: September 27th, 11:59pm EST.**

- **Google Doc Link:** **https://docs.google.com/document/d/1RMVPXrxFsWeWzznQ93QwOcRgF3mpgJXNWynUV06jBMA/edit?usp=sharing**

- **Timeline for the Project** *(*changes are subject to course schedule and may be notified to students during the term)*

Tentative Deadline	Project Phase	Deliverables
October 13th	**Phase 1:** Project Proposal (**8%**)	A **5-min** presentation of the state-of-the-art review of applied ML algorithm in your selected topic + Proposed ML solutions
December 8th	**Phase 2:** Final presentation (**10%**)	A **10-min** presentation of the code implementation workflow, outcomes, and final remarks

December 14th	**Phase 3:** Final project report (**12%**)	The final course project report should comply with the **IEEE conference format** (double-column), and it should be **at least 5 pages long**. The state-of-the-art review (from phase 1) should be included in the final report. The report should include abstract, introduction, literature review, an overview of the selected ML algorithm, implementation results, conclusion, and references.

- **Potential Topics for Selection**
  1. **Signal/Image processing:** audio signal enhancement, high-resolution image generation, image-to-image translation
  2. **Communication system:** Wireless channel prediction, cognitive radio for spectrum sensing, resource prediction or allocation in wireless networks
  3. **Smart grid:** load forecasting for data centers, renewable energy generation forecasting, load forecasting for residential areas, anomaly detection in smart grid
  4. **Robotics and embedded system:** real-time object recognition, energy-efficient edge computing for robotic tasks, predictive maintenance for robots
  5. **Health science and biomedical engineering:** electrocardiogram (ECG) data analysis, activity recognition with wearable sensor data

Note that the above are just some examples of the potential topics. You can also select a topic of your interests, as long as it falls within the definition of "**Electrical and Computer Engineering**". **For the code implementation**, if you are conducting a code implementation from a paper, the paper should be of high quality, and it should be ACM/IEEE indexed. For code implementation without references, the instructor and the TA will evaluate the technical contribution and feasibility and provide feedback to students.

- **Evaluation Criteria**

Your project will be evaluated based on the following criteria:

- **Clarity and Completeness:** Are your reports and presentations clear, well-structured, and easy to follow? Have you addressed all aspects of the project?
- **Technical Merit:** What is the technical depth of your chosen methodologies and implementations? Does your approach demonstrate a deep understanding of the selected topic?
- **Results and Analysis:** Are your results well-presented and thoroughly analyzed? Have you critically evaluated the performance of your models?

- **Project Report:** Are the project organized and well-written without typo? Does the project reflect what was presented in two presentations?

Sample papers for interested topics:

1. Signal/Image processing: audio signal enhancement, high-resolution image generation, image-to-image translation

[1] T. Tarasiewicz, J. Nalepa and M. Kawulok, "A Graph Neural Network For Multiple-Image Super-Resolution," *2021 IEEE International Conference on Image Processing (ICIP)*, Anchorage, AK, USA, 2021, pp. 1824-1828, doi: 10.1109/ICIP42928.2021.9506070.

[2] Q. Bao, B. Gang, W. Yang, J. Zhou and Q. Liao, "Attention-Driven Graph Neural Network for Deep Face Super-Resolution," in *IEEE Transactions on Image Processing*, vol. 31, pp. 6455-6470, 2022, doi: 10.1109/TIP.2022.3212311.

2. Communication system: Wireless channel prediction, cognitive radio for spectrum sensing, resource prediction or allocation in wireless networks

[1] W. Wang, C. Zhou, H. He, W. Wu, W. Zhuang and X. Shen, "Cellular Traffic Load Prediction with LSTM and Gaussian Process Regression," *ICC 2020 - 2020 IEEE International Conference on Communications (ICC)*, Dublin, Ireland, 2020, pp. 1-6, doi: 10.1109/ICC40277.2020.9148738.

[2] J. M. J. Huttunen, D. Korpi and M. Honkala, "DeepTx: Deep Learning Beamforming With Channel Prediction," in *IEEE Transactions on Wireless Communications*, vol. 22, no. 3, pp. 1855-1867, March 2023, doi: 10.1109/TWC.2022.3207055.

3. Smart grid: load forecasting for data centers, renewable energy generation forecasting, load forecasting for residential areas, anomaly detection in smart grid

[1] M. Mughees, Y. Li, Y. Chen and Y. R. Li, "Short-Term Load Forecasting for AI-Data Center," *2025 IEEE Power & Energy Society General Meeting (PESGM)*, Austin, TX, USA, 2025, pp. 1-5, doi: 10.1109/PESGM52009.2025.11225317.

[2] Y. Zhang, J. Wang and B. Chen, "Detecting False Data Injection Attacks in Smart Grids: A Semi-Supervised Deep Learning Approach," in *IEEE Transactions on Smart Grid*, vol. 12, no. 1, pp. 623-634, Jan. 2021, doi: 10.1109/TSG.2020.3010510.

4. Robotics and embedded system: real-time object recognition, energy-efficient edge computing for robotic tasks, predictive maintenance for robots

[1] X. Wang and H. Guo, "Mobility-Aware Computation Offloading for Swarm Robotics using Deep Reinforcement Learning," *2021 IEEE 18th Annual Consumer Communications & Networking Conference (CCNC)*, Las Vegas, NV, USA, 2021, pp. 1-4, doi: 10.1109/CCNC49032.2021.9369456.

[2] Chen C, Liu C, Wang T, et al. Compound fault diagnosis for industrial robots based on dual-transformer networks[J]. Journal of Manufacturing Systems, 2023, 66: 163-178.

5. Health science and biomedical engineering: electrocardiogram (ECG) data analysis, activity recognition with wearable sensor data

[1] E. Merdjanovska and A. Rashkovska, "Patient-Specific Heartbeat Classification in Single-Lead ECG using Convolutional Neural Network," *2021 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, Mexico, 2021, pp. 932-936, doi: 10.1109/EMBC46164.2021.9630366.

[2] A. Procházka *et al*., "Deep Learning for Accelerometric Data Assessment and Ataxic Gait Monitoring," in *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 29, pp. 360-367, 2021, doi: 10.1109/TNSRE.2021.3051093.